

No Profitable Trade: From KKT to Revealed Preference

A Brainstorming Paper and Research Agenda

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Status. This document is a disciplined brainstorming paper, not a novelty claim and not a finished submission. It separates established mathematics from interpretive synthesis, candidate diagnostics, and open research questions. The organizing phrase “no profitable trade” is deliberately economic; it is not asserted to be new terminology. Any eventual paper must add a systematic novelty search, sharpen the statistical objects, and choose one principal theorem rather than pursue every extension at once.

Abstract

This note develops a common language for constrained optimization and revealed preference. For a proposed choice, exact regret asks whether another feasible bundle delivers higher utility; its first-order counterpart is the support gap obtained by maximizing the current utility gradient over the opportunity set. For a concave objective, zero support gap is equivalent to global optimality. On a standard budget the gap has a closed form: it decomposes into unspent-income loss and commodity-specific entry losses, reproducing all KKT conditions—including corners— without requiring multipliers at the discovery stage. Multipliers are recovered afterward as dual certificates.

Across repeated budgets, the same geometry leads to Afriat inequalities. Observation-specific rescalings of prices serve as supporting utility gradients; GARP is the observable no-profitable-cycle condition left after utility levels and marginal utilities of income are eliminated. A parallel blocking gap extends the idea to asymmetric preference functions and coherent coalitional multi-utility representations, with an exact warning: no blocking is generally weaker than preference maximization unless skew symmetry is imposed. The note closes with proposed stochastic-choice and continuum-good extensions, a formal-verification program, and an explicit ledger distinguishing classical results from potentially new work.

One proposed conceptual spine

$$\begin{array}{c} \text{no improving feasible direction} \implies \text{no profitable trade} \\ \Downarrow \\ \text{no profitable revealed-preference cycle} \end{array}$$

The first arrow is optimization within one opportunity set. The second is an integrability question across many opportunity sets. They should be connected, but never conflated.

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1 The question and the levels of information

The proposed program starts from one question:

Given a proposed or observed choice, can one exhibit a feasible trade, a revealed-preference cycle, a blocking coalition, or a stochastic separating trial that demonstrates an improvement?

The answer depends on what is observed. It is useful to distinguish four levels.

- L1. Known objective and one feasible set.** We can compute exact regret and a first-order no-profitable-trade gap.
- L2. Known objective and many feasible sets.** Every chosen point must pass the one-set test, and the gradients must arise from the same objective.
- L3. Unknown objective and finite demand observations.** Supporting gradients and utility levels are latent. Afriat's inequalities ask whether such objects exist; GARP is the corresponding observable cycle condition.
- L4. Population choice probabilities.** A distribution of deterministic rational types is latent. Convex separation replaces a single revealed-preference graph.

This hierarchy prevents two recurring mistakes. First, a numerical first-order gap is not directly observable when utility is unknown. Second, a revealed-preference cycle is not literally an arbitrage or a sequence of trades that the consumer could execute through time. It is a logical inconsistency among the preference comparisons implied by choices, unless an additional market mechanism gives the cycle a trading interpretation.

2 One opportunity set: exact regret and the NPT gap

Let $C \subset \mathbb{R}^L$ be nonempty, compact, and convex, and let u be differentiable on a neighborhood of C . For $x \in C$, define the *exact utility regret*

$$\mathcal{R}_u(x \mid C) := \max_{y \in C} \{u(y) - u(x)\}, \quad (1)$$

and the *linearized no-profitable-trade gap*

$$\mathcal{G}_u(x \mid C) := \max_{y \in C} \langle \nabla u(x), y - x \rangle = \sigma_C(\nabla u(x)) - \langle \nabla u(x), x \rangle, \quad (2)$$

where $\sigma_C(g) := \max_{y \in C} \langle g, y \rangle$ is the support function of C . The maximizing y in (2) is the best trade according to current marginal utilities. In optimization, this object is the variational-inequality or Frank–Wolfe gap [4, 5]; the proposed contribution is to make its economic content central and to connect it systematically to revealed preference.

Proposition 2.1 (Exact regret is bounded by the NPT gap). *Suppose u is concave and differentiable. For every $x \in C$,*

$$0 \leq \mathcal{R}_u(x \mid C) \leq \mathcal{G}_u(x \mid C).$$

Moreover, the following statements are equivalent:

- (i) $x \in \arg \max_{y \in C} u(y)$;
- (ii) $\mathcal{R}_u(x \mid C) = 0$;
- (iii) $\mathcal{G}_u(x \mid C) = 0$;

(iv) $\langle \nabla u(x), y - x \rangle \leq 0$ for every $y \in C$.

Proof. Because $x \in C$, both maxima are at least zero. Concavity gives

$$u(y) - u(x) \leq \langle \nabla u(x), y - x \rangle \quad \text{for every } y \in C,$$

and hence $\mathcal{R}_u \leq \mathcal{G}_u$. Thus $\mathcal{G}_u = 0$ implies global optimality. Conversely, if x is globally optimal, then for each $y \in C$ the function $\varphi(t) = u(x + t(y - x))$ is maximized at $t = 0$ over $[0, 1]$. Therefore $\varphi'(0) = \langle \nabla u(x), y - x \rangle \leq 0$. Taking the maximum over y proves $\mathcal{G}_u = 0$. $\square$

The proof makes the economics transparent. Concavity converts the linearized value of the best feasible trade into a valid upper bound on the value of the best actual deviation. Without concavity, $\mathcal{G}_u = 0$ remains a first-order stationarity condition but need not establish a global maximum.

Remark 2.2 (A positive gap certifies a direction, not necessarily a full jump). If $\mathcal{G}_u(x | C) > 0$ and y^ℓ solves the support problem, then $d = y^\ell - x$ has positive directional derivative. Convexity of C and differentiability of u imply that $x + \gamma d$ is feasible and improves utility for every sufficiently small $\gamma > 0$. Curvature may nevertheless make the full jump from x to y^ℓ unattractive. A line search or a sufficiently small step converts the linear certificate into an actual profitable trade.

2.1 Nonsmooth utilities

For a concave function, let $\partial^+ u(x)$ denote its superdifferential. Given $g \in \partial^+ u(x)$, define

$$\mathcal{G}(x; g, C) := \sigma_C(g) - \langle g, x \rangle.$$

The supergradient inequality immediately yields

$$\mathcal{R}_u(x | C) \leq \mathcal{G}(x; g, C). \quad (3)$$

Thus any $g \in \partial^+ u(x)$ with $\mathcal{G}(x; g, C) = 0$ is a global optimality certificate. This version is the right bridge to Afriat's piecewise-affine rationalizing utility, which is typically nondifferentiable.

2.2 Ordinal content and its limit

Zero is robust, magnitude is not. If $\tilde{u} = \phi \circ u$ and ϕ is differentiable with $\phi'(u(x)) > 0$, then

$$\nabla \tilde{u}(x) = \phi'(u(x)) \nabla u(x), \quad \mathcal{G}_{\tilde{u}}(x | C) = \phi'(u(x)) \mathcal{G}_u(x | C).$$

Consequently, the statement “there is no profitable first-order trade” is locally ordinally invariant, but its utility-unit magnitude is not. A numerical NPT gap must therefore be normalized or tied to a specified cardinal representation before it is interpreted as a welfare quantity.

3 A single budget: a closed form and KKT recovery

Consider the standard budget

$$B(p, m) := \{y \in \mathbb{R}_+^L : p \cdot y \leq m\}, \quad p \in \mathbb{R}_{++}^L, \quad m > 0.$$

Fix a feasible x and write $g = \nabla u(x)$. Define the largest positive marginal utility per dollar

$$\rho(x) := \max \left\{ 0, \max_{1 \leq i \leq L} \frac{g_i}{p_i} \right\}. \quad (4)$$

Theorem 3.1 (One-budget NPT formula). *For every feasible x ,*

$$\mathcal{G}_u(x \mid B(p, m)) = m\rho(x) - g \cdot x \quad (5)$$

$$= \rho(x) [m - p \cdot x] + \sum_{i=1}^L p_i x_i \left[\rho(x) - \frac{g_i}{p_i} \right]. \quad (6)$$

If u is concave, x is globally optimal if and only if this expression is zero. Equivalently,

$$m - p \cdot x \geq 0, \quad \rho \geq 0, \quad \rho(m - p \cdot x) = 0, \quad (7)$$

$$g_i - \rho p_i \leq 0, \quad x_i(g_i - \rho p_i) = 0 \quad (i = 1, \dots, L). \quad (8)$$

Proof. The linear support problem is

$$\max_{y \geq 0, p \cdot y \leq m} g \cdot y.$$

If all $g_i \leq 0$, choosing $y = 0$ gives value zero. Otherwise, all expenditure is placed on a good attaining the highest positive ratio g_i/p_i , so the value is $m\rho$. Subtracting $g \cdot x$ yields (5); adding and subtracting $\rho p \cdot x$ yields (6). Every term in the latter is nonnegative by feasibility and the definition of ρ . Hence the sum is zero exactly when all complementary products in (7)–(8) vanish. Global optimality follows from Proposition 2.1. $\square$

The formula turns the KKT conditions into an accounting identity. Set

$$\lambda := \rho, \quad \mu_i := \rho p_i - g_i \geq 0. \quad (9)$$

Then $g - \lambda p + \mu = 0$, while (6) is precisely

$$\lambda(m - p \cdot x) + \mu \cdot x.$$

For a generic candidate, (ρ, μ) is the optimal dual certificate for the *linearized* support problem. Thus the NPT calculation can discover and screen an allocation without first guessing multipliers. When the gap is zero, concavity makes x optimal for the original utility problem and the same pair becomes an original-problem KKT certificate: ρ is a shadow-income supergradient and μ_i is the exclusion margin for a zero-consumption good.

Corollary 3.2 (Interior equalization and corner screening). *If $\rho > 0$, zero NPT gap requires the budget to bind. Every consumed good satisfies*

$$\frac{\partial_i u(x)}{p_i} = \rho,$$

whereas every excluded good satisfies

$$\frac{\partial_i u(x)}{p_i} \leq \rho.$$

An excluded good with a strictly larger ratio is an immediate profitable-entry certificate.

Example 3.3 (A corner detected before writing the Lagrangian). Let

$$u(x_1, x_2) = a_1 \log(1 + x_1) + a_2 x_2, \quad a_1, a_2 > 0,$$

and consider the candidate $\bar{x} = (0, m/p_2)$. At this candidate,

$$g = (a_1, a_2), \quad \rho = \max \left\{ \frac{a_1}{p_1}, \frac{a_2}{p_2} \right\},$$

so

$$\mathcal{G}_u(\bar{x} \mid B) = m \left[\frac{a_1}{p_1} - \frac{a_2}{p_2} \right]_+.$$

The all-good-2 corner is globally optimal exactly when $a_1/p_1 \leq a_2/p_2$. If the inequality fails, the gap identifies good 1 as the profitable entrant. The actual utility gain from reoptimization is no larger than this linearized bound.

Example 3.4 (Cobb–Douglas as an exact zero-gap audit). For $u(x) = \alpha \log x_1 + (1 - \alpha) \log x_2$, the familiar demand

$$x_1^* = \frac{\alpha m}{p_1}, \quad x_2^* = \frac{(1 - \alpha)m}{p_2}$$

has

$$\frac{\partial_1 u(x^*)}{p_1} = \frac{\partial_2 u(x^*)}{p_2} = \frac{1}{m}.$$

Hence $\rho = 1/m$, $g \cdot x^* = 1$, and $m\rho - g \cdot x^* = 0$. This one-line calculation simultaneously checks budget exhaustion, equalized bang per buck, and global optimality. Here utility is understood on the positive orthant (or as an extended-value utility equal to $-\infty$ on the axes); the gradient audit is applied at the strictly positive optimizer.

3.1 A candidate scale-free diagnostic

When $g \geq 0$ and $\rho > 0$, define

$$\widehat{\mathcal{G}}_u(x \mid B) := \frac{\mathcal{G}_u(x \mid B)}{m\rho} = 1 - \frac{g \cdot x}{m\rho} \in [0, 1]. \quad (10)$$

This quantity is invariant to multiplying g by a positive scalar and therefore to a smooth increasing transformation of utility with strictly positive derivative at $u(x)$. It is a promising classroom diagnostic: it reports the share of maximal current marginal value lost to slack income and misallocation. It is *not yet* a revealed-preference efficiency index or a cardinal welfare measure. Its relationship to the critical cost efficiency index and other data-only measures is an open question, not a theorem asserted here.

4 Many linear constraints: multipliers as the dual of trade search

Let

$$C = \{y \in \mathbb{R}_+^L : Ay \leq b\}$$

be nonempty and compact. For a proposed $x \in C$ and a marginal-value vector g , linear programming duality gives the exact identity

$$\mathcal{G}(x; g, C) = \max_{y \in C} g \cdot (y - x) \quad (11)$$

$$= \min_{\lambda \geq 0: A^\top \lambda \geq g} \{b^\top \lambda - g^\top x\} \quad (12)$$

$$= \min_{\lambda \geq 0: A^\top \lambda \geq g} \{\lambda^\top (b - Ax) + (A^\top \lambda - g)^\top x\}. \quad (13)$$

Writing $\mu = A^\top \lambda - g \geq 0$, a zero gap is equivalent to the existence of prices of the constraints such that

$$g - A^\top \lambda + \mu = 0, \quad \lambda_j (b_j - A_j x) = 0, \quad \mu_i x_i = 0.$$

These are the KKT conditions under the maximization convention $L = u(x) + \lambda^\top (b - Ax) + \mu^\top x$ [7].

This duality clarifies what a “multiplier-free KKT” can and cannot mean. The primal support problem avoids multipliers as a method of discovery: search directly for the best feasible trade. The dual variables remain mathematically valuable because they compress the proof that no trade is profitable. Eliminating them from the student’s first calculation does not eliminate duality.

4.1 Sparse local tests

Let $T_C(x)$ be the tangent cone of C at x . The first-order NPT condition is

$$g \cdot d \leq 0 \quad \text{for every } d \in T_C(x).$$

For a polyhedron, it is enough to check a generating set of extreme rays of the tangent cone. This suggests a sparse “elementary trade” implementation: pairwise swaps for a single budget, network cycles for flow constraints, and circuit directions for a general matrix A . The underlying convex geometry is standard; the research task is to identify economically meaningful generators and minimal certificates for important classes of models.

5 From NPT to revealed preference

Consider a finite demand dataset

$$\mathcal{D} = \{(p^t, x^t)\}_{t \in \mathcal{T}}, \quad p^t \in \mathbb{R}_{++}^L, \quad x^t \in \mathbb{R}_+^L \setminus \{0\},$$

with observed wealth $m^t = p^t \cdot x^t$ and budget $B_t = B(p^t, m^t)$. Bundle x^t is directly revealed preferred to x^s when

$$p^t \cdot x^s \leq p^t \cdot x^t.$$

GARP rules out a directed affordability cycle containing only weak edges and at least one strict edge. Equivalently, it rules out x^t being indirectly revealed preferred to x^s while x^s is strictly directly revealed preferred to x^t [1, 13].

5.1 The graph intuition

Suppose first that one common, locally nonsatiated utility function rationalizes the data. Every outgoing affordability edge $t \rightarrow s$ implies

$$u(x^t) \geq u(x^s).$$

Transitivity of numerical utility propagates this inequality along every directed path. A strict affordability edge, together with local nonsatiation, produces a strict preference comparison. Therefore a strict directed cycle is impossible.

This yields the useful but carefully qualified slogan:

NPT is no improving edge within a known opportunity set. GARP is no inconsistent cycle across observed opportunity sets when the objective is unknown.

5.2 Afriat inequalities as compatible local certificates

Afriat's theorem states that GARP is equivalent to the existence of numbers $U_t \in \mathbb{R}$ and $\lambda_t > 0$ satisfying

$$U_s \leq U_t + \lambda_t p^t \cdot (x^s - x^t) \quad \text{for all } s, t \in \mathcal{T}. \quad (14)$$

The piecewise-affine function

$$u_A(x) := \min_{t \in \mathcal{T}} \{U_t + \lambda_t p^t \cdot (x - x^t)\} \quad (15)$$

is concave, continuous, strictly increasing, and rationalizes the data. At x^t the t th plane is active, so

$$q_t := \lambda_t p^t \in \partial^+ u_A(x^t). \quad (16)$$

Moreover,

$$\mathcal{G}(x^t; q_t, B_t) = \max_{y \in B_t} q_t \cdot (y - x^t) = 0.$$

Thus each observation receives a zero-NPT certificate, while inequalities (14) ensure that all certificates are supporting covectors of one concave potential.

Theorem 5.1 (Afriat–NPT bridge: a restatement of known results). *For a finite Walrasian demand dataset, the following statements are equivalent:*

- (i) *the dataset satisfies GARP;*
- (ii) *the Afriat inequalities (14) have a solution with $\lambda_t > 0$;*
- (iii) *the data are rationalizable by a continuous, strictly increasing, concave utility function;*
- (iv) *there exist a concave strictly increasing function u and positive rescaled prices $q_t = \lambda_t p^t$ such that*

$$q_t \in \partial^+ u(x^t) \quad \text{and} \quad \mathcal{G}(x^t; q_t, B_t) = 0 \quad \text{for every } t;$$

- (v) *there are scalars $\lambda_t > 0$ such that the graph $\{(x^t, q_t)\}_{t \in \mathcal{T}}$ is cyclically supermonotone:*

$$\sum_{k=1}^K q_{t_k} \cdot (x^{t_{k+1}} - x^{t_k}) \geq 0, \quad t_{K+1} = t_1, \quad (17)$$

for every finite cycle of observations.

Statements (i)–(iii) are Afriat’s theorem. The equivalence with (iv) is the supporting- hyperplane interpretation above. Summing the supergradient inequalities around a cycle gives (v). Conversely, for fixed q_t the Afriat system is the complete directed difference-constraint system

$$U_s - U_t \leq q_t \cdot (x^s - x^t).$$

Such a system is feasible exactly when every directed cycle has nonnegative total weight. The resulting U_t and the minimum-of-affine construction (15) produce the concave potential. Finally, $q_t = \lambda_t p^t$ and $p^t \cdot x^t = m^t$ imply that q_t supports B_t at x^t , hence its local support gap is zero. This is the finite-data form of Rockafellar’s cyclic-monotonicity construction [11]. The word *supermonotone* records the sign convention for concave supergradients; $-q_t$ belongs to the usual convex cyclically monotone formulation.

Interpretation of λ_t . Price is not itself marginal utility. The vector $\lambda_t p^t$ is the supporting utility covector, and λ_t is the observation-specific marginal utility of income. Raw price vectors generally need not be cyclically monotone because income effects and cardinal utility scales vary across observations. In quasilinear environments with an interior numeraire, the normalization often fixes $\lambda_t = 1$; this is why unscaled cyclic monotonicity plays a direct role in quasilinear rationalizability [10].

5.3 Why corners do not break the bridge

Suppose a differentiable concave u rationalizes the observations and the KKT conditions at observation t are

$$\nabla u(x^t) = \lambda_t p^t - \mu_t, \quad \mu_t \geq 0, \quad \mu_t \cdot x^t = 0. \quad (18)$$

For every observed x^s , concavity yields

$$u(x^s) - u(x^t) \leq \nabla u(x^t) \cdot (x^s - x^t) \quad (19)$$

$$= \lambda_t p^t \cdot (x^s - x^t) - \mu_t \cdot x^s + \mu_t \cdot x^t \quad (20)$$

$$\leq \lambda_t p^t \cdot (x^s - x^t). \quad (21)$$

This is the Afriat inequality. The commodity-specific nonnegativity multipliers disappear from the observable inequality because $-\mu_t \cdot x^s \leq 0$ and $\mu_t \cdot x^t = 0$. Corners therefore strengthen rather than obstruct the certificate.

6 Preference functions and coalitional blocking

Let $r : X \times X \rightarrow \mathbb{R}$ be a reflexive preference function normalized by $r(x, x) = 0$, where $r(x, y) \geq 0$ means that x is weakly preferred to y . Standard utility is the special case $r(x, y) = u(x) - u(y)$. For a feasible set B and $x \in B$, define two gaps:

$$\mathcal{P}_r(x \mid B) := \sup_{y \in B} [-r(x, y)]_+, \quad \text{preference-maximization gap,} \quad (22)$$

$$\mathcal{B}_r(x \mid B) := \sup_{y \in B} r(y, x), \quad \text{blocking gap.} \quad (23)$$

Reflexivity makes both gaps nonnegative. The first is zero exactly when $r(x, y) \geq 0$ for every $y \in B$. The second is zero exactly when no feasible y satisfies $r(y, x) > 0$.

Proposition 6.1 (Maximization, core, and skew symmetry). *Suppose r is asymmetric in the sense that $r(x, y) \geq 0$ implies $r(y, x) \leq 0$. Then*

$$\mathcal{P}_r(x \mid B) = 0 \implies \mathcal{B}_r(x \mid B) = 0.$$

If r is skew-symmetric, $r(y, x) = -r(x, y)$, then

$$\mathcal{P}_r(x \mid B) = \mathcal{B}_r(x \mid B),$$

so preference maximization and no blocking are equivalent.

Proof. If $r(x, y) \geq 0$ for every feasible y , asymmetry implies $r(y, x) \leq 0$. Since $r(x, x) = 0$, the supremum in (23) is zero. Under skew symmetry,

$$\sup_{y \in B} [-r(x, y)]_+ = \sup_{y \in B} [r(y, x)]_+ = \sup_{y \in B} r(y, x),$$

where the last equality again uses the available zero at $y = x$. □

This proposition records an essential caveat: with a merely asymmetric preference function, a core point need not be a preference maximizer. Treating the blocking gap as exact regret is valid under skew symmetry, but generally false without it.

6.1 Coalitional multi-utility

For a transparent finite formulation, let Ω be a nonempty finite collection of nonempty finite sets U of utility functions. A CMU preference function has the form

$$r_\Omega(y, x) = \max_{U \in \Omega} \min_{u \in U} \{u(y) - u(x)\}. \quad (24)$$

Its blocking gap is

$$\mathcal{B}_{r_\Omega}(x \mid B) = \sup_{y \in B} \max_{U \in \Omega} \min_{u \in U} \{u(y) - u(x)\}. \quad (25)$$

On a compact budget with continuous component utilities, zero means that no coalition U and feasible bundle y make every utility in U strictly favor y over x . The compact CMU formulation replaces finiteness by the corresponding compactness and continuity hypotheses.

Aguiar, Hjertstrand, Serrano, and Evren characterize WGARP through asymmetric preference functions and coherent CMU representations [2]. Their Section 6 studies exactly the associated core problem

$$\max_{y \in B} r_\Omega(y, \bar{x}).$$

For such a coherent CMU, at $\bar{x} = x^*$ a preference maximizer solves this problem; the converse holds under skew symmetry. Thus the NPT language is not a replacement for that theory. It is a proposed unifying interpretation and a route to quantitative, computational, and formal extensions.

For classical utility, (25) collapses to

$$\mathcal{B}_r(x \mid B) = \sup_{y \in B} \{u(y) - u(x)\} = \mathcal{R}_u(x \mid B).$$

The hierarchy can therefore be stated without ambiguity:

$$\mathcal{P}_{r_\Omega}(x \mid B) = 0 \implies_{\text{asymmetry}} \mathcal{B}_{r_\Omega}(x \mid B) = 0, \quad \mathcal{P}_{r_\Omega} = 0 \iff_{\text{skew symmetry}} \mathcal{B}_{r_\Omega} = 0.$$

For the singleton classical representation $r(x, y) = u(x) - u(y)$, skew symmetry holds and both objects equal exact utility regret.

The magnitudes of $\mathcal{P}_r$ and $\mathcal{B}_r$ depend on the cardinal representation r and on normalizations of the component utilities. Their zero sets are the robust objects. Any empirical “distance from WGARP” based on these gaps must specify an admissible normalization class.

7 Stochastic choice: a separating-trial gap

For a finite stochastic-choice design and a fixed population whose distribution of latent preference types is stable across the observed choice sets, stack choice probabilities in a vector $\pi \in \mathbb{R}^J$. Let $a^1, \dots, a^R$ denote the incidence vectors generated by all deterministic rational choice types admitted by the design, and set

$$P := \text{conv}\{a^1, \dots, a^R\}.$$

Random-utility rationalizability is equivalent to $\pi \in P$ in this finite formulation [8, 6]. Given a norm $\|\cdot\|$ and its dual $\|\cdot\|_*$, define

$$\mathcal{G}_{\text{stoch}}(\pi) := \max_{\|w\|_* \leq 1} \{w \cdot \pi - \sigma_P(w)\}, \quad \sigma_P(w) = \max_r w \cdot a^r. \quad (26)$$

Convex duality gives

$$\mathcal{G}_{\text{stoch}}(\pi) = \inf_{q \in P} \|\pi - q\|. \quad (27)$$

Hence the gap is zero exactly under rationalizability. When it is positive, an optimizing w is a unit-dual-norm normal vector to a separating hyperplane: a stochastic revealed-preference trial on which the observed probabilities outperform every deterministic rational type. This is the probabilistic counterpart of a profitable trade.

Equation (27) is standard convex separation, not a claimed new characterization. The research opportunities are economic and statistical:

- choose a norm and normalization with a transparent behavioral interpretation;
- obtain sparse separating trials that can be explained to a student or decision maker;
- derive sampling theory and critical values for the maximized gap;
- study robustness to measurement error, building on stochastic revealed-preference methods such as Aguiar and Kashaev [3];
- extend the construction from GARP-rational types to WGARP/CMU types.

8 A continuum of differentiated goods

The continuum-good extension is motivated by product environments in which the finite list of commodities is itself an approximation. Let $(\Omega, \mathcal{F}, \nu)$ be a nontrivial, σ -finite measure space of product varieties, and let $p : \Omega \rightarrow (0, \infty)$ be measurable. The commodity space is the weighted cone $L_+^1(p \, d\nu)$. For a finite positive expenditure level $0 < m < \infty$, define

$$B(p, m) = \left\{ x \in L_+^1(p \, d\nu) : \int_{\Omega} p(\omega)x(\omega) \, d\nu(\omega) \leq m \right\}.$$

Suppose the derivative of utility at x is the continuous linear functional

$$Du(x)[d] = \int_{\Omega} g(\omega)d(\omega) \, d\nu(\omega),$$

where g is measurable and $g/p \in L^\infty(\nu)$. Set

$$\rho := \operatorname{ess\,sup}_{\omega \in \Omega} \max \left\{ 0, \frac{g(\omega)}{p(\omega)} \right\}. \quad (28)$$

Redefining the support gap with a supremum on this noncompact density space, the support calculation gives

$$\begin{aligned} \mathcal{G}(x; g, B) &:= \sup_{y \in B(p, m)} \int_{\Omega} g(\omega)[y(\omega) - x(\omega)] \, d\nu(\omega) \\ &= m\rho - \int_{\Omega} g(\omega)x(\omega) \, d\nu(\omega) \end{aligned} \quad (29)$$

$$= \rho \left[m - \int_{\Omega} p(\omega)x(\omega) \, d\nu(\omega) \right] + \int_{\Omega} p(\omega)x(\omega) \left[\rho - \frac{g(\omega)}{p(\omega)} \right] \, d\nu(\omega). \quad (30)$$

The proof is the finite-good argument after the change of variable $z(\omega) = p(\omega)y(\omega)$. The support value is m times the essential supremum of the positive part of g/p . Sigma-finiteness permits expenditure to be concentrated on finite-measure subsets on which this ratio approaches its essential supremum. For $m > 0$ and $\rho > 0$, a density maximizer exists exactly when $\{\omega : g(\omega)/p(\omega) = \rho\}$ has positive measure; otherwise only a concentrating sequence approaches the supremum. If $\rho = 0$, the zero density attains the support value.

The functional-analytic support identity is standard. A serious continuum consumer theory must still settle the economically important issues:

- C1. Choice space.** Densities, finite measures, and indivisible product choices imply different feasible spaces and dual objects.
- C2. Differentiability.** The appropriate Fréchet, Gâteaux, or directional derivative depends on the topology and on whether utility includes taste for variety or bundle interactions.
- C3. Attainment.** Bliss-point or discrete-choice consumers may choose atoms rather than densities, changing both compactness and the support problem.
- C4. Neighborhood substitution.** Cross-price effects should be aggregated over sets of nearby products, not reported as an unmanageably large matrix indexed by individual varieties.
- C5. Discretization.** A theorem is needed showing when finite catalogs and their NPT certificates converge to the continuum model.

Research question 8.1 (Neighborhood revealed preference). Let products occupy a metric characteristic space (Ω, d) and let $N_h(\omega)$ be an h -neighborhood. Can one define revealed substitution kernels and neighborhood cross-price elasticities that (i) are identified from discretized demand, (ii) converge as the catalog becomes dense, and (iii) admit sparse NPT witnesses localized to economically interpretable product neighborhoods?

9 A formal-verification program

The framework is well suited to proof certificates because failure and success both admit finite witnesses in the finite-dimensional cases.

1. **One budget.** Verify the identity

$$\mathcal{G} = m\rho - g \cdot x = \rho(m - p \cdot x) + \sum_i x_i(\rho p_i - g_i)$$

and conclude zero from nonnegative terms and complementarity.

2. **General polyhedra.** Verify primal feasibility, dual feasibility, stationarity, complementary products, and the supporting-hyperplane global bound.
3. **GARP failure.** A finite list of observations, affordability inequalities, and one strict inequality is a complete cycle certificate.
4. **GARP success.** Exact rational numbers (U_t, λ_t) satisfying all Afriat inequalities certify rationalizability; the minimum-of-affine construction supplies the mathematical bridge.
5. **Stochastic failure.** A rational vector w whose separating inequality is positive certifies nonmembership in the rational-type polytope.

Lean should certify the smallest exact algebraic claim that matters; it should not be described as having established novelty, prose quality, or an infinite-dimensional analytic theorem that has not been formalized. A practical repository would separate: (a) data certificates; (b) algebraic verification; and (c) the analytic assumptions connecting those certificates to economic meaning.

10 What is known, what is synthesis, and what remains open

Status	Object	Assessment
Established	Concavity, variational inequalities, and the support/Frank–Wolfe gap	Zero gap characterizes a concave maximum; the gap bounds objective regret.
Established	KKT and linear-programming duality	Multipliers are dual certificates for the absence of a profitable linearized trade.
Established	Afriat inequalities, GARP, and concave rationalization	These equivalences are classical; the NPT vocabulary is a restatement and teaching bridge.

Continued on the next page

Status	Object	Assessment
Established	Cyclic monotonicity and integrability	Rockafellar supplies the mathematical potential theorem; positive rescaled prices, not generally raw prices, are the relevant covectors.
Established	WGARP and coherent CMU rationalization	Aguiar et al. provide the rationalization and core problem; no-blocking is weaker than maximization without skew symmetry.
Established	Finite random-utility polytopes and separation	Rationalizability is polytope/cone membership; a support inequality separates failures.
Interpretive synthesis	“NPT $\rightarrow$ no profitable cycle $\rightarrow$ rationalizability”	Potentially valuable as a unified exposition, but not itself a theorem of novelty.
Candidate diagnostic	Normalized local NPT gap (10)	Locally invariant to utility rescaling; its empirical and welfare content remains to be proved.
Research agenda	Quantitative CMU and k -acyclic blocking gaps	Requires a principled normalization and theorems relating gap size to observable violations.
Research agenda	Sparse stochastic separating trials	Requires statistical theory, computational guarantees, and comparison with existing ARSP tests.
Research agenda	Continuum goods and neighborhood substitution	Requires a choice-space model, existence and discretization results, and economically meaningful cross-price objects.
Research agenda	Machine-checked certificate library	Finite algebraic certificates are feasible; analytic bridges must be stated separately and honestly.

11 A focused paper sequence

Trying to establish every extension in one paper would obscure the central insight. A credible sequence works backward from sharply stated deliverables.

Paper 1: the finite deterministic bridge

The first paper should center on one theorem or diagnostic beyond the classical equivalences. A disciplined target is:

Construct a normalized, representation-aware NPT diagnostic for finite consumer data; characterize its primal profitable-trade and dual price-support forms; and establish its exact relationship to Afriat efficiency and sparse revealed-preference cycles.

The literature review must determine whether this object is already implicit in approximate revealed-preference, variational-inequality, or inverse-optimization work.

Paper 2: CMU and bounded cycles

Define normalized blocking gaps for coherent CMU representations, determine what can be identified without fixing a cardinal representation, and relate the minimal gap to WGARP and k -acyclicity. The core/maximization distinction must remain explicit throughout.

Paper 3: stochastic certificates

Develop sparse separating trials, finite-sample inference, and robust versions that allow measurement error. The contribution should be statistical interpretability or computational sparsity, not the already-known convex separation characterization.

Paper 4: product neighborhoods in a continuum catalog

Specify a characteristics-based continuum model, prove convergence from finite catalogs, and define neighborhood substitution objects. This paper should connect the billion-product motivation to a mathematically disciplined limit experiment rather than merely replacing sums with integrals.

12 Implications for the textbook

Even before the research agenda is resolved, the established portion supplies a clean teaching sequence:

1. Utility maximization introduces the model and exact regret.
2. The NPT gap shows why marginal conditions, corners, and global concavity belong together.
3. KKT multipliers appear as recovered prices certifying that every feasible trade loses.
4. Repeated observations turn local support into an integrability problem.
5. Afriat and GARP solve the inverse problem: whether local certificates can be glued into one utility function.
6. WGARP and CMU weaken the gluing requirement from all cycles to pairwise consistency and coalitional no blocking.
7. Stochastic choice replaces a chosen bundle by a distribution and a profitable trade by a separating trial.

This ordering makes revealed preference an implication and inverse problem of utility maximization, not an apparently unrelated collection of axioms. It also permits one notation throughout: x for a proposed choice, B for its opportunity set, g or q for a supporting marginal-value covector, and a gap for the best admissible challenge.

13 Conclusion

The core mathematics is simple enough to be memorable:

$$\text{optimality} \iff \max_{y \in C} \langle \nabla u(x), y - x \rangle = 0$$

for differentiable concave utility. On a budget, the maximum has a closed form and decomposes into the familiar KKT complementary products. Across budgets, positive rescalings of price vectors are supporting utility covectors; Afriat inequalities test whether they can be integrated into a single concave potential, and GARP is the observable cycle restriction. For nontransitive preference functions, a blocking gap survives, but its equivalence to maximization requires skew symmetry.

The potential research contribution is not to rename KKT, Frank–Wolfe, Afriat, or Rockafellar. It is to build an economics-native theory of certificates—local trades, global cycles, blocking coalitions, stochastic trials, and continuum neighborhoods—with transparent normalization, sparse witnesses, and formal verification. The next step is to select one quantitative theorem for the first paper and subject it to a dedicated novelty audit.

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